title: "Savitribai Phule Pune University --- Engineering Mathematics-I (2024 Pattern)"
code: "BSC-101-BES"
pattern: "2024"
semester: "I"
total_marks: 70
duration: "2½ Hours"

Seat No. ____________

SAVITRIBAI PHULE PUNE UNIVERSITY

Engineering Mathematics-I (BSC-101-BES)

2024 Credit Pattern | Semester I

Total Marks: 70 | Time: 2½ Hours

Instructions:

1. Q.1 is compulsory.
2. Attempt Q.2 or Q.3, Q.4 or Q.5, Q.6 or Q.7, Q.8 or Q.9, Q.10 or Q.11.
3. Neat diagrams must be drawn wherever necessary.
4. Assume suitable data, if necessary.
5. Use of electronic pocket calculator is allowed.

Q1) Write correct option for the following. [10]

a) Stationary point for $f(x,y) = x^2 + y^2 + 6x + 12$ is: i) $(1,0)$ ii) $(3,0)$ iii) $(-3,0)$ iv) $(0,0)$ [2]

b) In the expansion of $\log(1+e^x)$ in powers of x, coefficient of x is: i) 1 ii) 1/2 iii) 1/4 iv) 0 [2]

c) If $u = \frac{x^2 + y^2}{x + y}$, then $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} =$: i) u ii) $2u$ iii) 0 iv) $3u$ [2]

d) Rank of matrix $\begin{bmatrix} 2 & -1 & -1 \\ 1 & 2 & 1 \\ 4 & -7 & -5 \end{bmatrix}$ is: i) 3 ii) 2 iii) 1 iv) 0 [2]

e) If A is $n\times n$ with eigen values $\lambda_1,\lambda_2,\ldots,\lambda_n$, then $\det(A) =$: i) $\sum\lambda_i$ ii) $\prod\lambda_i$ iii) 0 iv) $n\cdot\prod\lambda_i$ [1]

f) The quadratic form $3x^2 + 2y^2 + 3z^2 - 2xy - 2yz$ is: i) Positive definite ii) Negative definite iii) Indefinite iv) Semi-definite [1]

Unit I --- Single Variable Calculus (14 Marks)

Q2) Attempt the following: [12]

a) Verify Rolle's theorem for $f(x) = (x-1)(x-2)(x-3)$ in $[1,3]$. [6]

b) Expand $e^{x}\sin x$ in powers of x using Maclaurin's theorem up to x^3 term. [6]

OR

Q3) Attempt the following: [12]

a) Find the Fourier series of $f(x) = x$ in $(-\pi,\pi)$. Deduce that $\sum_{n=1}^\infty \frac{(-1)^{n+1}}{n} = \frac{\pi}{4}$. [6]

b) Find the half-range sine series for $f(x) = x(\pi - x)$ in $(0,\pi)$. [6]

Unit II --- Multivariable Calculus (14 Marks)

Q4) Attempt the following: [12]

a) If $u = x^2y + y^2z + z^2x$, show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = (x+y+z)^2$. [6]

b) If $u = \tan^{-1}\left(\frac{x^3 + y^3}{x - y}\right)$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \sin 2u$. [6]

OR

Q5) Attempt the following: [12]

a) If $z = f(x,y)$ where $x = e^u\cos v$, $y = e^u\sin v$, prove that $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = e^{-2u}\left(\frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial v^2}\right)$. [6]

b) Find the total derivative of $u = \sin(xy)$ where $x = e^t$, $y = t^2$. [6]

Unit III --- Applications of Partial Differentiation (14 Marks)

Q6) Attempt the following: [12]

a) If $x = r\cos\theta$, $y = r\sin\theta$, find $\frac{\partial(x,y)}{\partial(r,\theta)}$ and $\frac{\partial(r,\theta)}{\partial(x,y)}$. [6]

b) Find the maximum and minimum values of $f(x,y) = x^3 + 3xy^2 - 15x - 12y$. [6]

OR

Q7) Attempt the following: [12]

a) Find the stationary values of $x^2 + y^2 + z^2$ subject to $x + y + z = 3a$ and $xy + yz + zx = 0$. [6]

b) The period of a simple pendulum is $T = 2\pi\sqrt{l/g}$. If l is increased by 2% and g decreased by 3%, find percentage change in T. [6]

Unit IV --- Linear Algebra (14 Marks)

Q8) Attempt the following: [12]

a) Find the rank of matrix $\begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & -2 \\ 3 & 6 & 3 & -7 \end{bmatrix}$. [6]

b) Solve the system: $x + y + z = 6$, $x - y + z = 2$, $2x + y - z = 1$ using matrix inversion method. [6]

OR

Q9) Attempt the following: [12]

a) Examine for consistency: $x + 2y + z = 2$, $2x - y - z = 2$, $4x - 7y - 5z = 2$. [6]

b) Examine linear dependence of vectors: $(1,2,-1,3)$, $(2,4,1,-2)$, $(3,6,3,-7)$. [6]

Unit V --- Eigen Values (14 Marks)

Q10) Attempt the following: [12]

a) Find eigen values and eigen vectors of $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$. [6]

b) Reduce the quadratic form $2x^2 + 2y^2 + 2z^2 - 2xy - 2yz - 2zx$ to canonical form by orthogonal transformation. Find its nature. [6]

OR

Q11) Attempt the following: [12]

a) Verify Cayley-Hamilton theorem for $A = \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix}$. Hence find A^{-1} and A^3. [6]

b) Diagonalize $A = \begin{bmatrix} 8 & -8 & -2 \\ 4 & -3 & -2 \\ 3 & -4 & 1 \end{bmatrix}$ by finding modal matrix P. [6]

*** END OF QUESTION PAPER ***